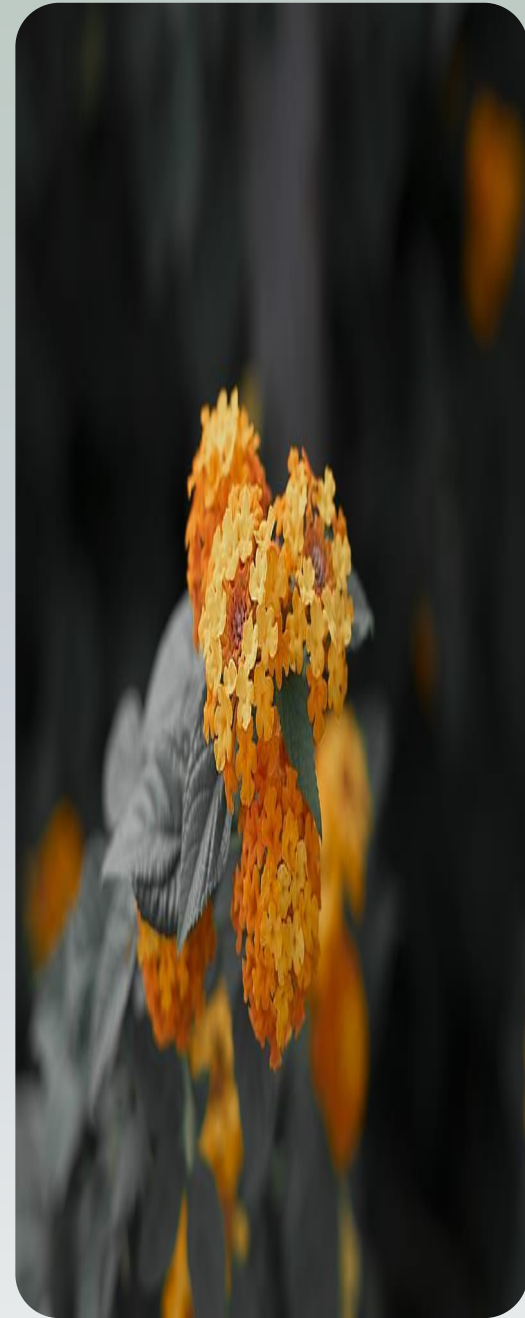


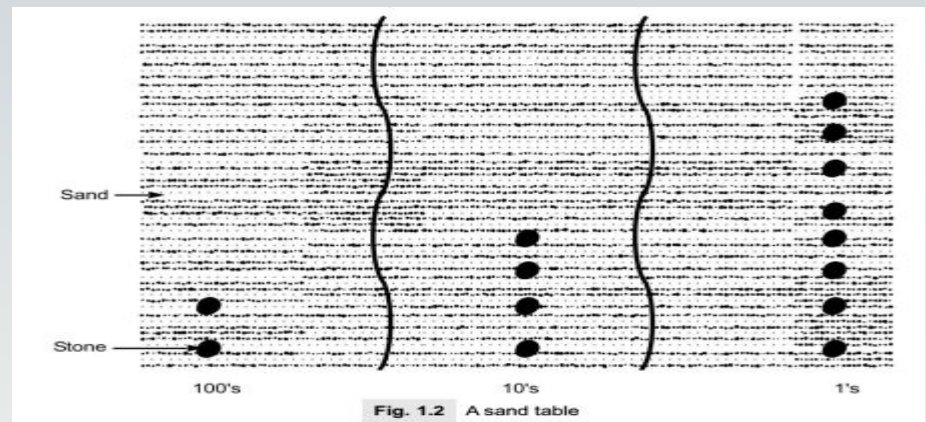
FIT Lecture # 3

Engr. Naveera Sami



EVOLUTION OF COMPUTERS

- Manual Computing Devices
- The idea of using stones for representing numbers and putting them at a place for performing simple calculations invented the device called sand table. A sand table was a device that arranged stones in three channels in the sand. Each channel could have a maximum of 10 stones. The addition operation was performed on this device by incrementing the count of right hand channel by one and by adding one stone in it. As soon as the right hand channel reached its maximum capacity, the stones were removed from that channel and one stone was added to the left hand channel. Figure 1.2 shows the idea of sand table used for the purpose of calculations.



Automated Computing Devices

- Charles Babbage, a professor of mathematics at the Cambridge University, made some worthwhile efforts towards automatic computing. He is also considered to be the father of modern computer. In 1812, Charles Babbage decided to automate the repeated series of steps needed in tabulating various functions, such as polynomial, logarithmic and trigonometric. In 1822, he presented a working model of his concept with the help of an automatic mechanical computing machine.
- He named the automatic mechanical computing machine as difference engine. In 1823, Babbage made it more automatic by providing the feature of printing the tabulated results. Babbage did not stop here and started working on developing the analytical engine.

GENERATIONS OF COMPUTERS

- Over the years, various computing devices were invented that enabled the people to solve different types of problems. All these computing devices can be classified into several generations. These generations refer to the phases of improvement made to different computing devices. The different phases of improvement made to computing devices resulted in a small, cheap, fast, reliable and productive computer. The technological development in the field of computers not only refers to the improvements made to the hardware technologies, but also the improvements made to the software technologies. The history of computer development is often discussed in terms of different generation of computers, as listed below.
- First generation computers
- Second generation computers
- Third generation computers
- Fourth generation computers
- Fifth generation computers

FIRST GENERATION COMPUTERS

The first generation computers were employed during the period 1940–1956

- These computers used the vacuum tubes technology for calculation as well as for storage and control purposes. Therefore, these computers were also known as vacuum tubes or thermionic valves based machines.
- Figure shows the vacuum tube used in first generation computers. A vacuum tube is made up of glass and contains filaments inside it. The filaments when heated generate electrons, which eventually help in the amplification and de amplification of electronic signals.
- The following are the two major advantages of first generation computer systems:
 - These computers were the fastest computing devices of their time.
 - These computers were able to execute complex mathematical problems in an efficient manner.

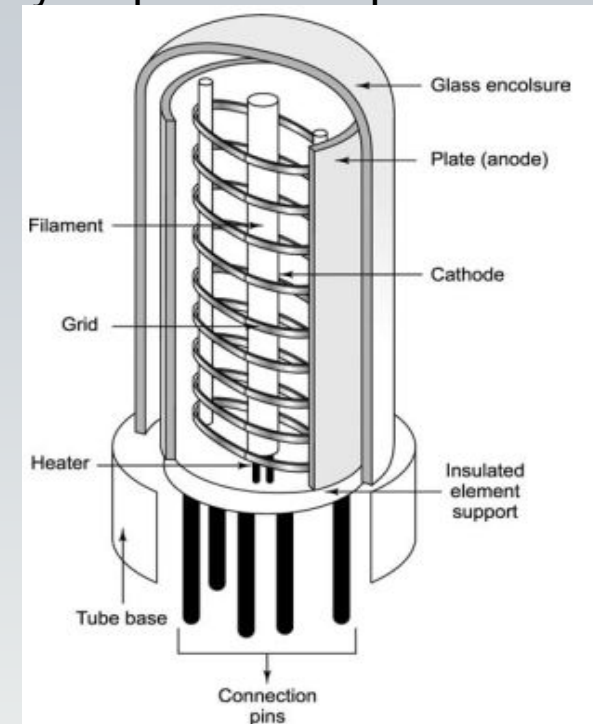


Fig. 1.5 A vacuum tube

FIRST GENERATION COMPUTERS

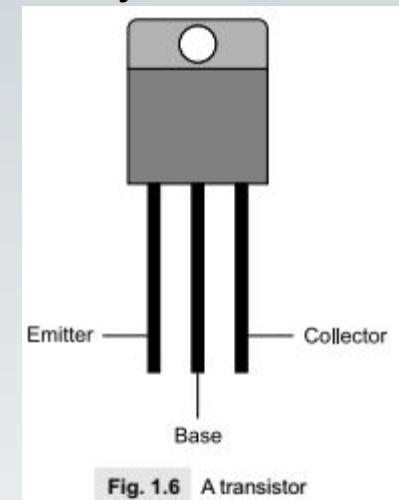
The first generation computers had many disadvantages associated with them. The following are some of the disadvantages of first generation computers:

- ❑ The functioning of these computers depended on the machine language. A machine language is a language in which all the values are represented in the form of 0s and 1s. Therefore, these computers were not very easy to program.
- ❑ They were generally designed as special-purpose computers. Therefore, they were not very flexible in running different types of applications.
- ❑ The use of vacuum tube technology made these computers very large and bulky. Due to their large size, it was not an easy task to install them properly.
- ❑ They were not easily transferable from one place to another due to their huge size and also required to be placed in cool places.
- ❑ They were single tasking because they could execute only one program at a time and hence, were not very productive.
- ❑ They generated huge amount of heat and hence were prone to hardware faults. Hence, they were not considered as reliable and required proper maintenance at regular intervals.

Second Generation Computers

The second generation computers were employed during the period 1956–1963

- The main characteristic of these computers was the use of transistors in place of vacuum tubes in building the basic logic circuits
- A transistor is a semiconductor device that is used to increase the power of the incoming signals by preserving the shape of the original signal. It has three connections, which are emitter (E), base (B) and collector (C). The base of transistor is the gate through which the signal, needed to be amplified, is sent. The signal sent through the base of the transistor is generally a small flow of electrons. Therefore, the base terminal also acts as the input gate for the transistor. The collector of the transistor is used to collect the amplified signal. The emitter of the transistor acts as the output gate for emitting the amplified signal to the external environment. Figure shows the transistor used to manufacture circuitry of second generation computers



Second Generation Computers

The following are the advantages of second generation computers:

- ❑ They were the fastest computing devices of their time.
- ❑ They were easy to program because of the use of assembly language.
- ❑ They could be transferred from one place to other very easily because they were small and light weight computing devices.
- ❑ They required very less power in carrying out their operations.
- ❑ They were more reliable as compared to first generation computers and hence, did not require maintenance at regular intervals of time.
- ❑ The following are the limitations of first generation computers:
- ❑ The input and output media for these computers were not improved to a considerable extent.
- ❑ They were required to be placed in air-conditioned places.
- ❑ The cost of these computers was very high and they were beyond the reach of home users.
- ❑ They were special-purpose computers and could execute only specific applications.

Third Generation Computers

The third generation computers were employed during the period 1964–1975.

- The major characteristic feature of third generation computer systems was the use of Integrated Circuits (ICs).
- The IC technology was also known as microelectronics technology. ICs are the circuits that combine various electronic components, such as transistors, resistors, capacitors, etc. onto a single small silicon chip. The first IC was developed by Jack Kilby and Robert Noyce in the year 1958.

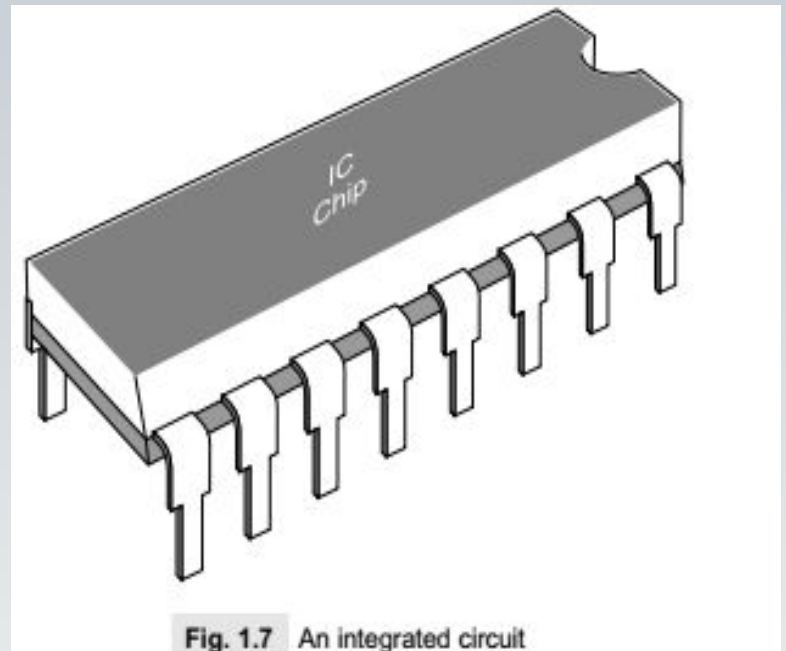


Fig. 1.7 An integrated circuit

Third Generation Computers

- ❑ ICs were superior to vacuum tubes and transistors in terms of cost and performance.
- ❑ The cost of ICs was very low and the performance was very high because all the electronic components were arranged very close to each other.
- ❑ They also required very low power for performing their operations. Therefore, the use of ICs in third generation computers made them smaller, faster, more efficient and more reliable than the first and second generation of computers.
- ❑ Some examples of third generation computers are NCR 395, B6500, IBM 370, PDP 11 and CDC 7600.

Third Generation Computers

The following are the merits of the third generation computers:

- They were the fastest computing devices as compared with first and second generation of computers. The computational time for these computers was also reduced to great extent. The computational time for these computers was usually measured in nanoseconds.
- They were very productive because of their small computational time.
- They were easily transportable from one place to another because of their small size.
- They used high-level languages. A high-level language is a computer programming language that is independent of the machine details. Hence, the programmer finds it very easy to use them and the programs written in these languages on one computer can be easily executed on some other computer.
- They could be installed very easily and required less space for their installation.
- They were able to execute any type of application, such as business and scientific applications. Hence, the third generation computers were also considered as general-purpose computers.
- They were more reliable and required less frequent maintenance schedules.

Third Generation Computers

- ❑ Some of the disadvantages of third generation computers are: Σ The storage capacity of these computers was still very small.
- ❑ The performance of these computers degraded while executing large applications, involving complex computations because of the small storage capacity.
- ❑ The cost of these computers was very high.
- ❑ They were still required to be placed in air-conditioned places.

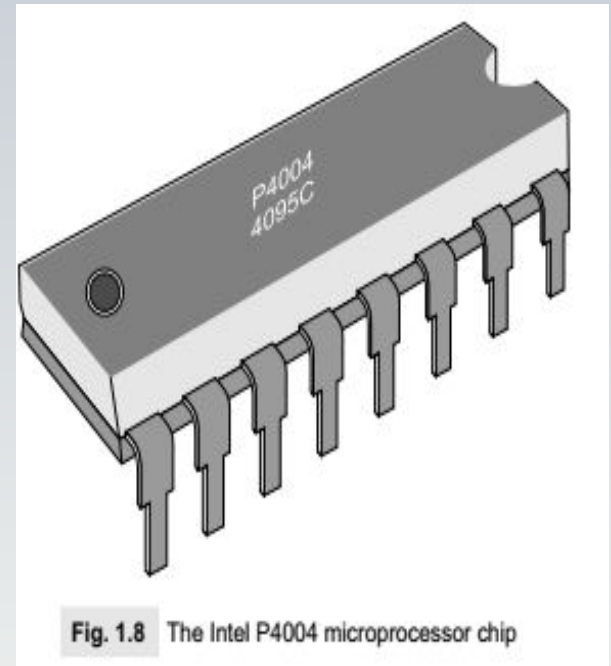
Fourth Generation Computers

The fourth generation computers were employed during 1975–1989.

- The invention of Large Scale Integration (LSI) technology and Very Large Scale Integration (VLSI) technology led to the development of fourth generation computers. However, these computers still used the IC technology to build the basic circuits.
- The LSI technology allowed thousands of transistors to be fitted onto one small silicon chip. On the other hand, the VLSI technology allowed hundreds of thousands of transistors to be fitted onto a single chip. As a result, the manufacturers were able to reduce the size of the computers and made them cheaper as compared to the other generation of computers
- The progress in LSI and VLSI technologies led to the development of microprocessor, which became the major characteristic feature of the fourth generation computers.
- A microprocessor incorporates various components of a computer—such as CPU, memory and Input / Output (I/O) controls—onto a single chip. The computers in this generation were designed to have a microprocessor, some additional storage chips and support circuitry. Figure 1.8 shows the Intel P4004 microprocessor chip developed in 1971. Some popular later microprocessors include Intel 386, Intel 486 and Pentium

Fourth Generation Computers

The term Personal Computer (PC) became known to the people during this era. The term PC refers to a computer that is designed to be used by an individual. Since the size and cost of the computer was decreased to a considerable extent in this period, people started using these computers for their personal work too. The storage technologies used in the fourth generation computers were also improved and they started using static and dynamic Random Access Memory (RAM). The advantage of using this type of memory was that it allowed the computers to access the stored information at a rapid pace and hence helped in increasing the productivity and performance of the computers. Some of the examples of fourth generation computers are IBM PC, IBM PC/AT, Apple and CRAY-1



Fourth Generation Computers

The use of LSI and VLSI technologies made the fourth generation computers small, cheap, compact and powerful. Apart from these technologies, the fourth generation also include the following developments:

- ❑ Development of Graphical User Interfaces (GUIs) ∑ Development of new operating systems
- ❑ Invention of various secondary storage and I/O devices ∑ Development of Local Area Network (LAN)
- ❑ Some of the advantages of fourth generation computers are as follows:
- ❑ The use of LSI, VLSI and semiconductor technologies made these computers very powerful in terms of their processing speed and access time.
- ❑ The storage capacity of these computers was very large and faster and hence they were very productive and highly optimized.
- ❑ They were highly reliable and required very less maintenance.
- ❑ They provided a user-friendly environment while working because of the development of GUIs and interactive I/O devices.

Fourth Generation Computers

- ❑ The programs written on these computers were highly portable because of the use of high-level
- ❑ languages.
- ❑ They were very versatile and suitable for every type of applications. ∑ They required very less power to operate.
- ❑ Some of the problems associated with fourth generation computers are as follows:
- ❑ The soldering of LSI and VLSI chips on the wiring board was not an easy task and required complicated technologies to bind these chips on the wiring board.
- ❑ The working of these computers is still dependent on the instructions given by the programmer.

Fifth Generation Computers

The different types of modern digital computers come under the categories of fifth generation computers.

- The fifth generation computers are based on the Ultra Large Scale Integration (ULSI) technology that allows almost ten million electronic components to be fabricated on one small chip.
- The ULSI technology helps in increasing the power and speed of the microprocessor chips and the capacity of primary and secondary storage devices to a great extent. As a result, the fifth generation computers are faster, cheaper and more efficient, as compared to the fourth generation computers.

Fifth Generation Computers

- Some of the improvements or developments made during this generation of computers are as follows:
- Development of various portable computers such as laptop, pocket computer, Personal Digital Assistant (PDA), etc. ∑ Development of Parallel Processors.
- Development of centralized computers called servers.
- Invention of optical disk technology.
- Invention of the Internet and its different services.

Fifth Generation Computers

- ❑ Some of the advantages of fifth generation computers are as follows:
- ❑ They are the fastest and powerful computers till date.
- ❑ They are able to execute a large number of applications at the same time and that too at a very high speed.
- ❑ The use of ULSI technology helps in decreasing the size of these computers to a large extent. Some of the fifth generation computers are so small in size that they can be used while traveling.
- ❑ The users of these computers find it very comfortable to use them because of the several additional multimedia features.
- ❑ They are versatile for communications and resource sharing.

Fifth Generation Computers

- ❑ The fifth generation computers are really enjoyed by their users because of the several advantages offered by them. However, the major disadvantage of the fifth generation computers is that they are not provided with an intelligent program that could guide them in performing different operations. Nowadays, scientists are making some serious efforts in this field and artificial intelligence and expert system applications are the results of these efforts.
- ❑ Figure 1.9 shows a tree of computer family that illustrates the area-wise developments during the last four decades and their contributions to the various generations of computers.

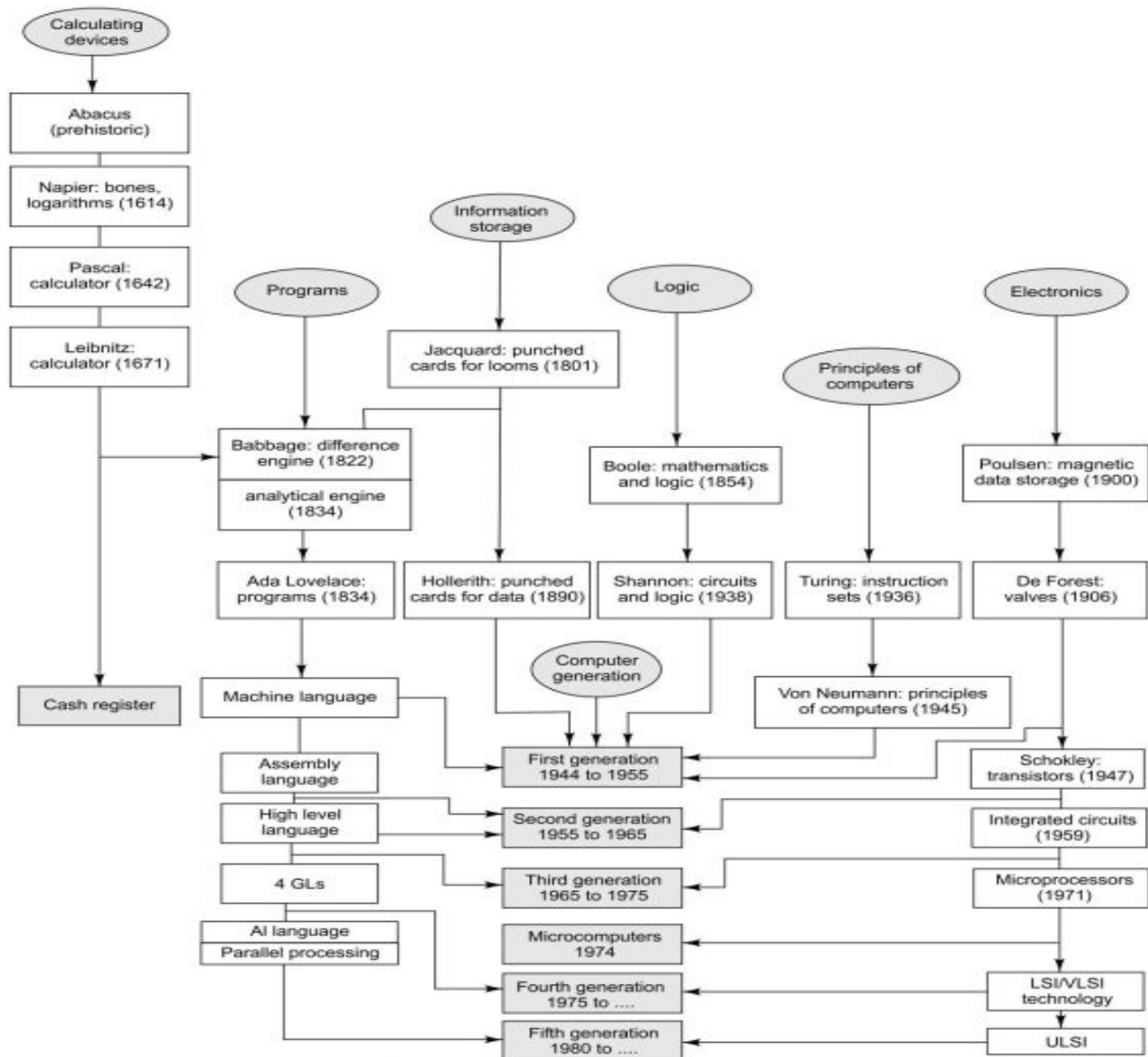


Fig. 1.9 Tree of computer family



Thank You